고려대학교 약학대학
KOREA UNIVERSITY COLLEGE OF PHARMACY

KU-The Future
110 years and beyond

Microbiology



Fall semester, 2026

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❖ General properties of antimicrobial agents

▶ Modes of action

· Five different modes of action of antimicrobial agent:

1. inhibition of cell wall synthesis
2. disruption of cell membrane function
3. inhibition of protein synthesis
4. inhibition of nucleic acid synthesis
5. action as antimetabolites

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❖ General properties of antimicrobial agents

▶ Modes of action

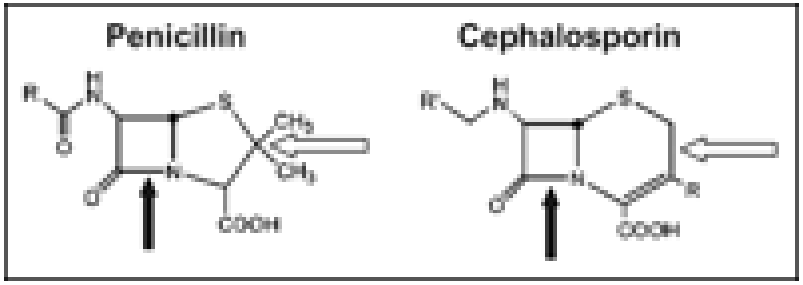
· Five different modes of action of antimicrobial agent:

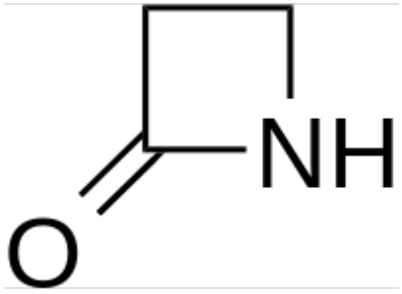
1. inhibition of cell wall synthesis
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4. inhibition of nucleic acid synthesis
5. action as antimetabolites

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❖ Beta-lactams

- Definition: antibiotics having the beta-lactam ring as a part of their core structure
- Nomenclature: In general, ending in -cillin or -penem or starting with ceph-, but not all cases
- Mode of action: inhibiting bacterial cell wall synthesis (nearly all of them)
- Resistance: by expressing the β-lactamase gene (capable of breaking the β-lactam ring) and producing new PBPs or changing PBPs

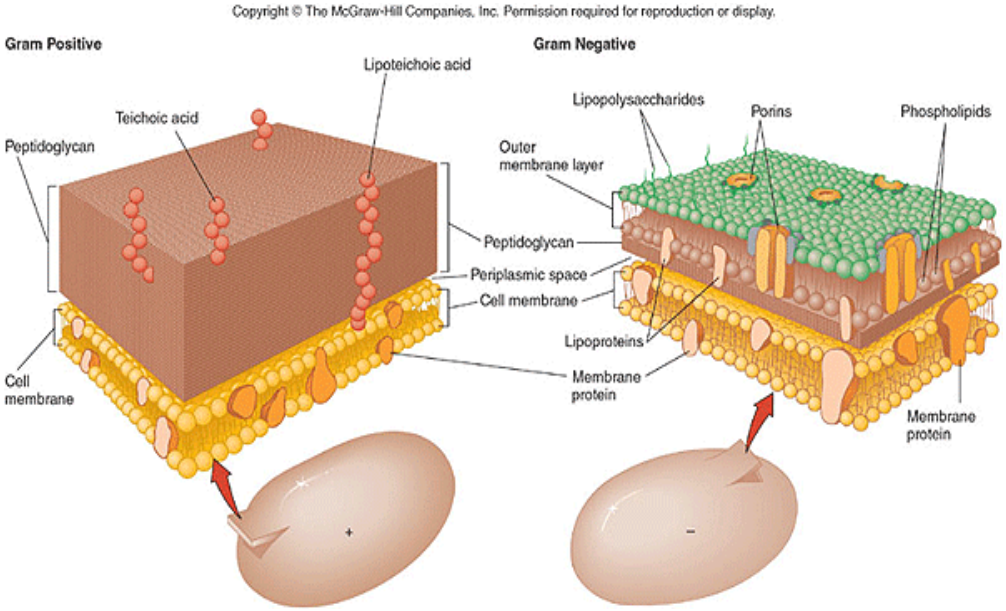


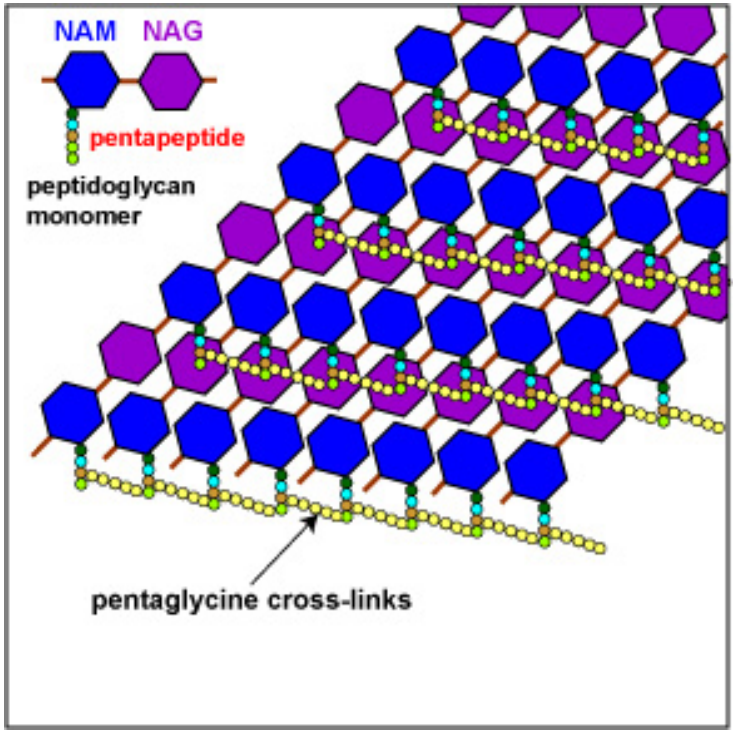


Beta-lactam

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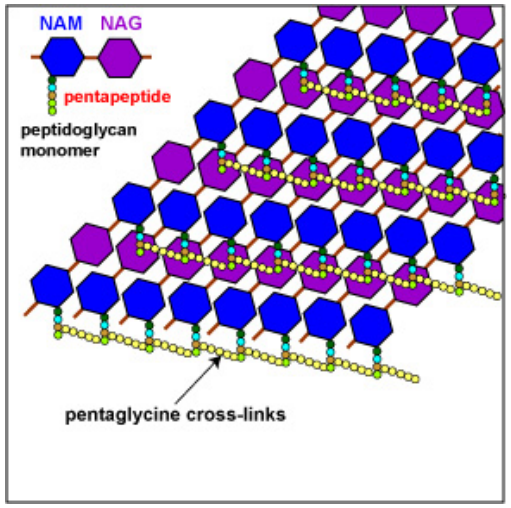
❖ Beta–lactams

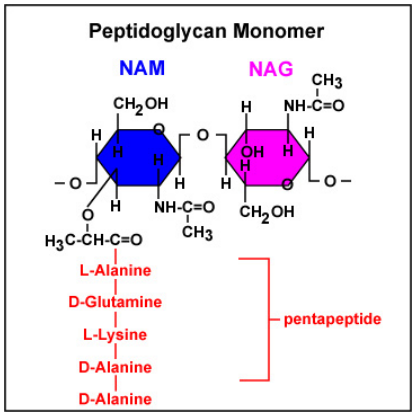




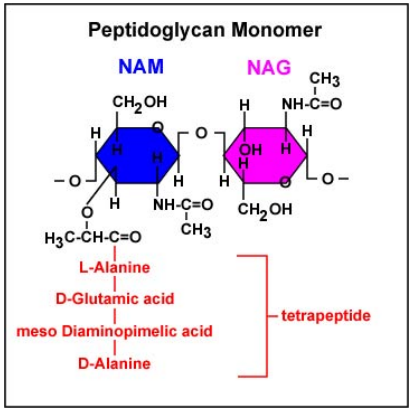
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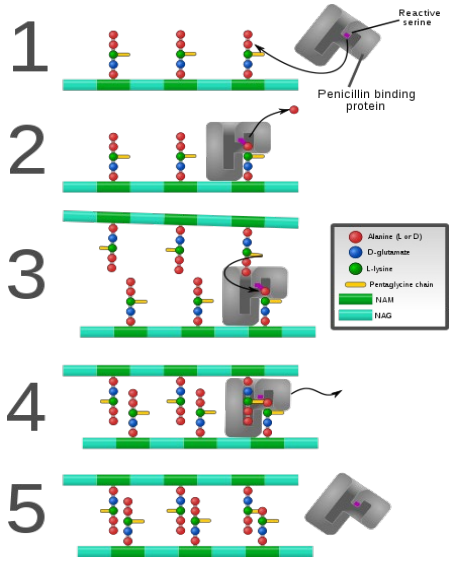
❖ Beta-lactams





Cross-linking





▶ Penicillin and the other β-lactam antibiotics:

- binding with a penicillin-binding protein (transglycosylase) (except: tabtoxinine-β-lactam)

→ inhibition of the cross-linking of the monomers

- classical: penicillins / cephalosporins
- non-classical: carbapenems / monobactams

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❖ Beta-lactams

· A few things in common:

- causing hypersensitivity reactions (called allergy): mild (skin rash, itch, fever, etc.) to severe (anaphylaxis; life-threatening allergic reaction)
- causing seizures by very high doses of any β-lactams
  → accumulation to toxic levels occur in patients having renal dysfunction
- lacking activity against atypical microorganisms: ex) *Mycoplasma pneumoniae*, etc.
- lacking activity against MRSA (nearly all of them)
  · MRSA: methicillin-resistant *Staphylococcus aureus*
  · methicillin: the first semisynthetic penicillin / effective against penicillin-resistant bacteria (PRSA)

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❖ Beta-lactams

▶ Penicillins

· a few things in common:

- very short half-lives (< 2 hours)
  → must be dosed multiples times per day
- like other β-lactams, causing hypersensitivity reactions
- being relatively poorly absorbed
  → can lead to diarrhea (in case of oral therapy)

· effective against:

- the most G (+) bacteria (e.g. the genera *Staphylococcus* and *Streptococcus*): bacterial pneumonia
- some of G (−) bacteria: e.g. the genus *Neisseria* and *Treponema pallidum*)

· retaining activity in urine, so that suitable for treating some urinary tract infections

· Thereafter, semisynthetic penicillins were developed.

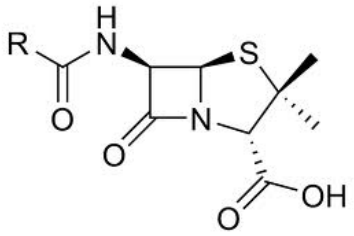
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❖ Beta-lactams

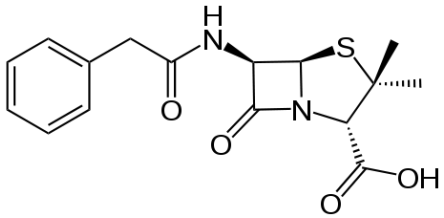
▶ Penicillins

· Natural penicillins:

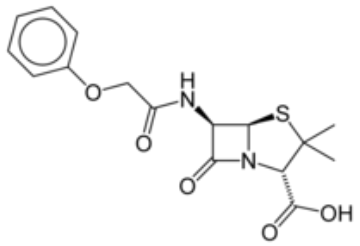
- extracted from the cultures of the *Penicillium notatum* (*currently, P. chrysogenum*)
- penicillin G (benzylpenicillin)
  - · the most frequently used natural penicillin
  - · administered parenterally (e.g. intramuscularly or intravenously)
  - · not administered orally: instability in gastric acid and weak intestinal absorption
- penicillin V (phenoxymethylpenicillin)
  - · oral form of penicillin G
  - · affected by food intake, so that recommended to be administered at 30 to 60 min before a meal



Penicillin
core structure



Penicillin G



Penicillin V

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❖ Beta-lactams

▶ Penicillins

· Natural penicillins:

– procaine benzylpenicillin:

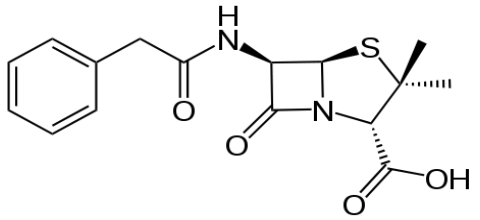
· a combination of benzylpenicillin and procaine
· procaine: local anaesthetic / allowing deep intramuscular injection

– benzathine benzylpenicillin:

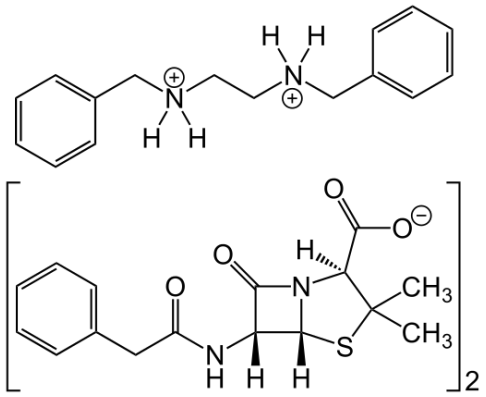
· benzathine: a diamine / stabilizing penicillins

– both above:

· slowly absorbed (by intramuscular injection) and released
· used where prolonged low concentration is required



Penicillin G

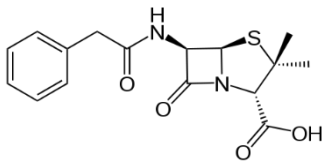


benzathine

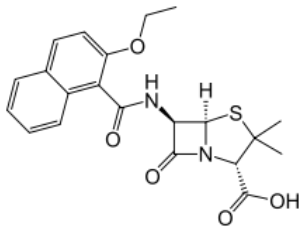
Classical beta-lactams (penicillins)

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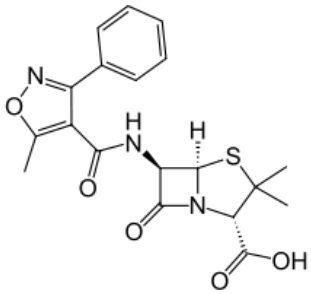
- ❖ Beta-lactams
- ▶ Penicillins
  - · Antistaphylococcal penicillins:
    - semisynthetic; resistant to β-lactamase
    - effective against penicillin-resistant staphylococci
    - methicillin, **nafcillin**, **oxacillin**, cloxacillin, **dicloxacillin**
    - short half-life, so that must be dosed frequently
      → causing phlebitis



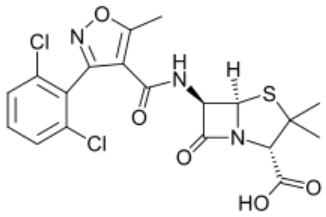
Penicillin G



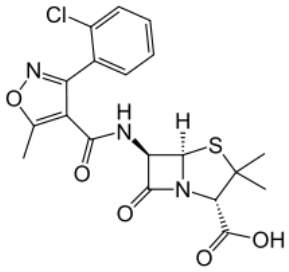
Nafcillin



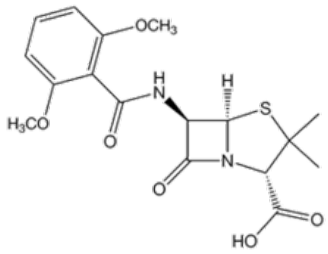
Oxacillin



Dicloxacillin



Cloxacillin

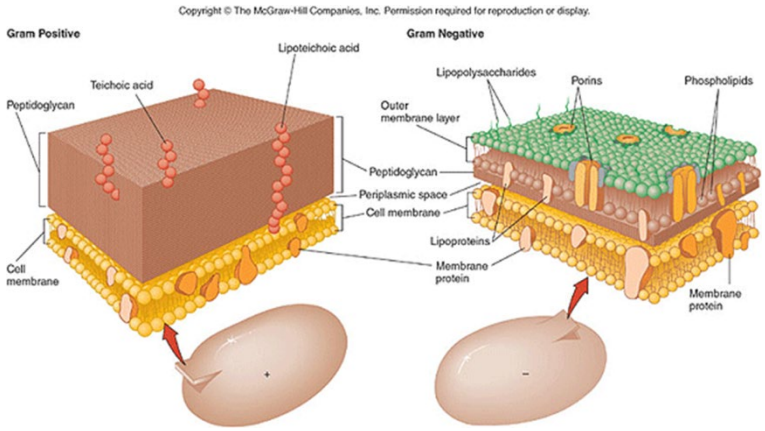


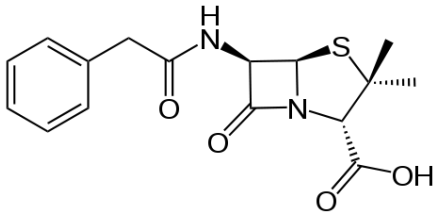
Methicillin

Classical beta-lactams (penicillins)

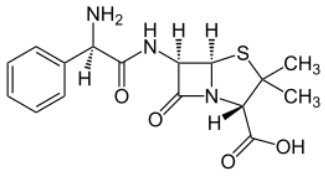
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- ❖ Beta-lactams
- ▶ Penicillins
  - · Aminopenicillins:
    - ampicillin, amoxicillin, ciclacillin, etc.
    - orally; extended spectrum β-lactams
      : increased coverage on G (-) bacteria,
      including enteric GNRs and *Haemophilus influenzae*
      · enteric GNRs: *Salmonella*, *Shigella*, *Escherichia*, *Vibrio*, *Yersinia*, etc.
    - more water-soluble (due to the introduction of an amino group)
      → passing through porins in the cell wall of G (-)
    - not effective against *Pseudomonas aeruginosa*

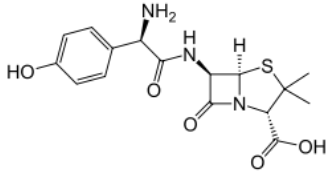




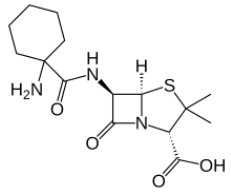
Penicillin G



Ampicillin



Amoxicillin



Ciclacillin

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❖ Beta-lactams

▶ Penicillins

· Aminopenicillins:

– not effective against *Pseudomonas aeruginosa* (녹농균)

* *Pseudomonas aeruginosa* (녹농균):

· the genus *Pseudomonas* : G (–) / environmental microbes
· the best studied species
· a common nosocomial (hospital-acquired) pathogen
  – causing inflammation and sepsis by colonization / infections in lung, urinary tract, kidney are fetal
· naturally resistant to most β-lactams, due to efflux pumps
  – efflux pumps: active transporters on the membrane / pumping out antibiotics before the antibiotics act

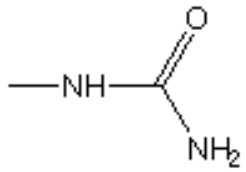
– still susceptible to β-lactamase; poor spectrum against *S. aureus* (PRSA and MRSA)

Classical beta-lactams (penicillins)

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❖ Beta-lactams

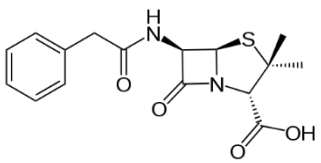
▶ Penicillins



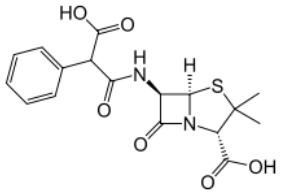
Ureido (NCON-)

· Antipseudomonal penicillins:

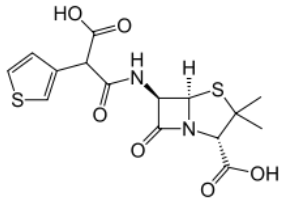
- extended spectrum: *P. aeruginosa*, etc.
- carbenicillin, ticarcillin, mezlocillin, piperacillin, azlocillin, etc.
  · carboxypenicillin: carbenicillin, ticarcillin
  · ureidopenicillins: mezlocillin, piperacillin and azlocillin
  → piperacillin: the most effective activity against *P. aeruginosa*
- still susceptible to β-lactamase: poor spectrum against *S. aureus* (PRSA and MRSA)



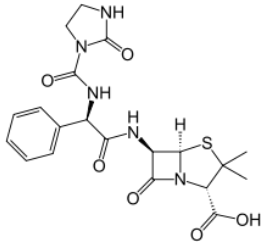
Penicillin G



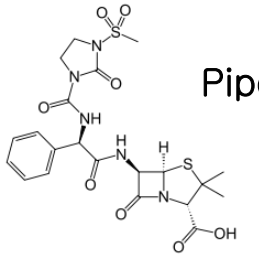
Carbenicillin



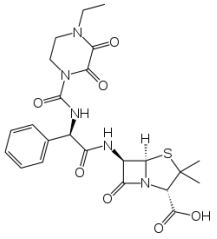
Ticarcillin



Azlocillin



Mezlocillin

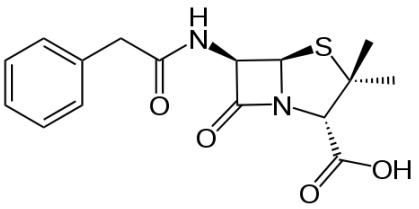


Piperacillin

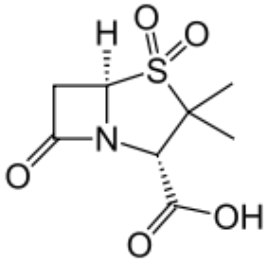
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❖ Beta-lactamase inhibitors

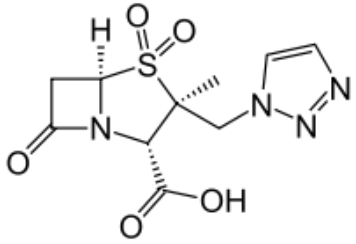
- definition: molecules used in conjunction with β-lactams to extend their spectrums of activity
- mimicking the structure of β-lactams, but have little or no antimicrobial activity on their own
- binding to β-lactamases irreversibly, so that preventing the β-lactamases from destroying any β-lactams
- co-administered with β-lactams to be effective
  : β-lactam/β-lactamase inhibitor combination
- sulbactam/ampicillin (Unasyn), tazobactam/piperacillin (Zosyn), clavulanate/amoxicillin (Augmentin) and /ticarcillin (Timentin)



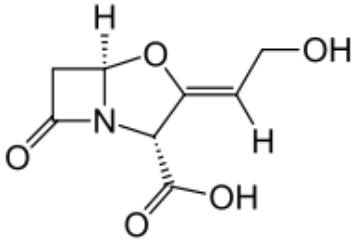
Penicillin G



Sulbactam



Tazobactam



Clavulanic acid

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❖ Beta-lactamase inhibitors

· The combination products are active only against the bacteria that the β-lactam in the combination has intrinsic activity against.

→ β-lactamase inhibitors restore activity, not add to it.

- Unasyn (ampicillin/sulbactam) is active against β-lactamase producing *E. coli*, because ampicillin alone is active against non-β-lactamase producing *E. coli*.
- It has no useful activity against *P. aeruginosa*, because ampicillin lacks activity against *P. aeruginosa*.

· not active against all β-lactamases; new β-lactamases are continually being discovered.